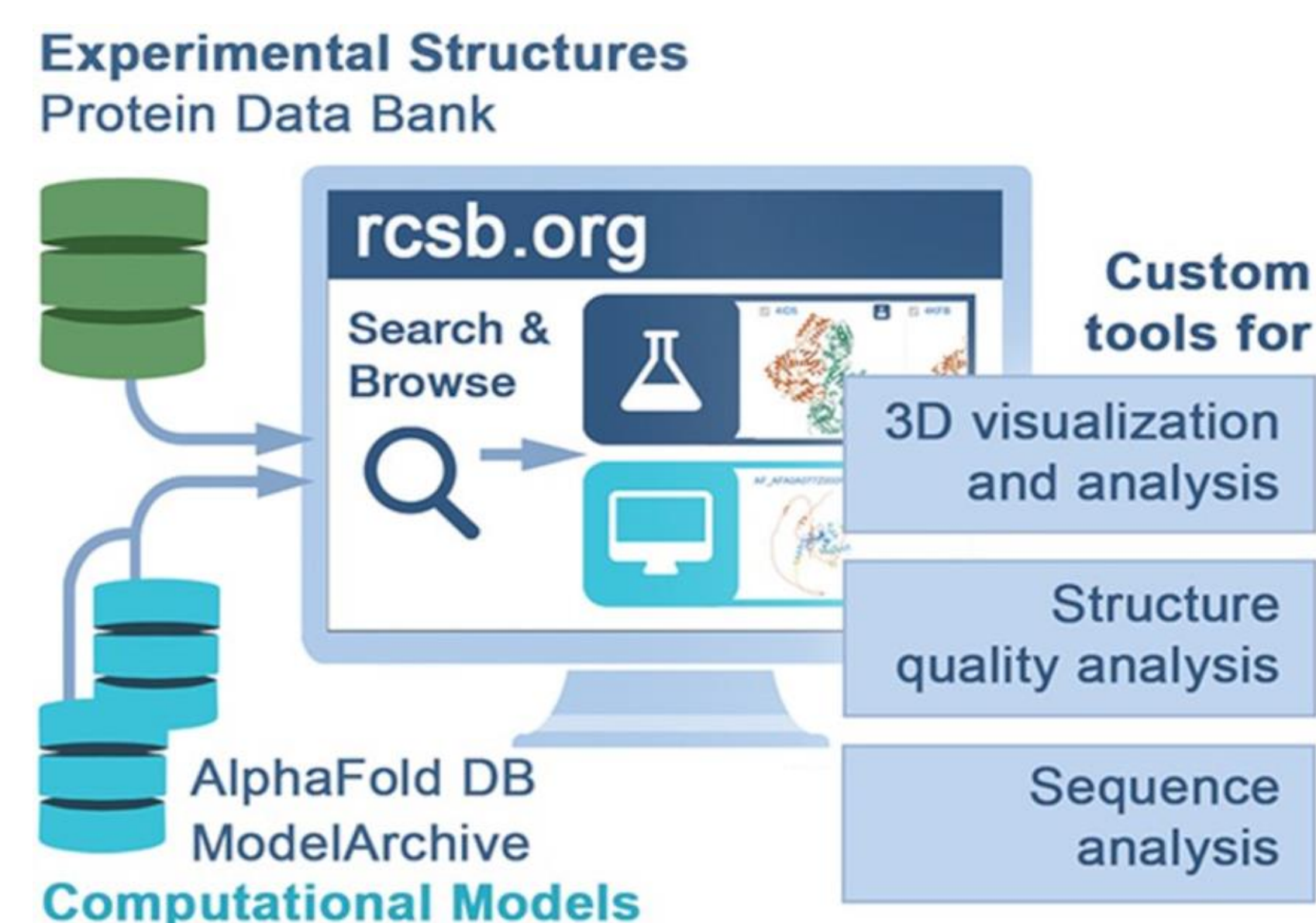


## Introduction

The Protein Data Bank (PDB), founded in 1971, has grown into a global archive with over 222,000 3D macromolecular structures of proteins, DNA, and RNA. It is today a leading global resource for experimental data central to scientific discovery. The RCSB.org portal facilitates data access, analysis, and visualization, supported by publicly available APIs.

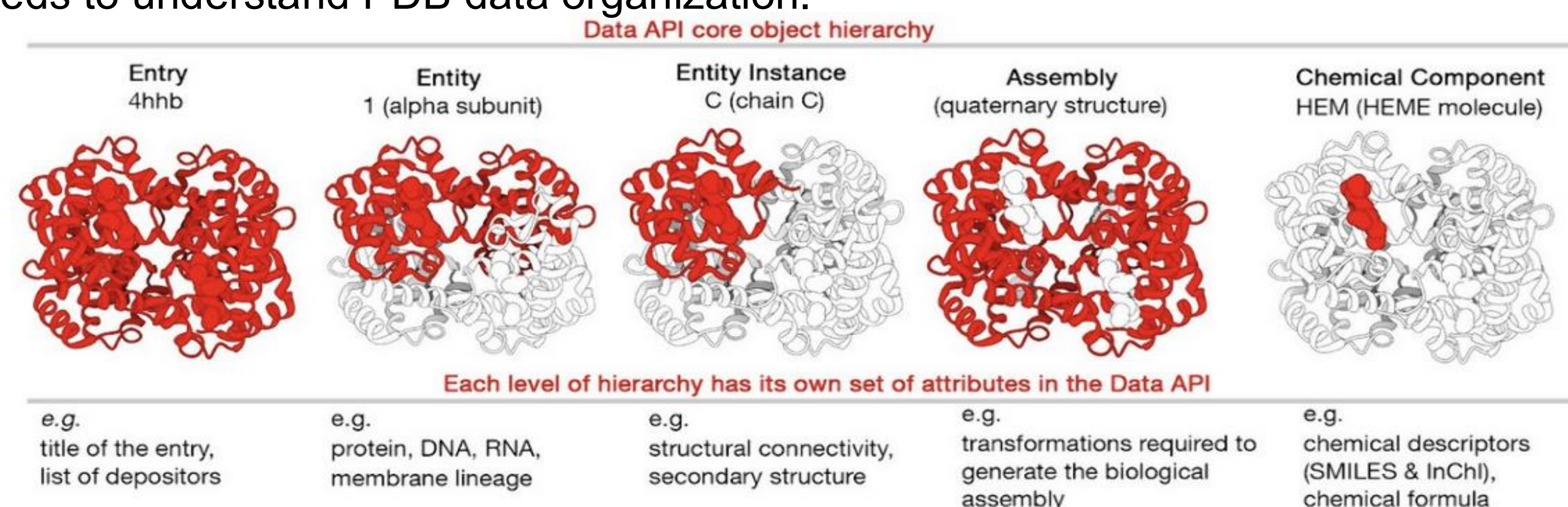


**Project Goal:** Develop a Python package that simplifies access to RCSB PDB Data API, enabling all users (even novice Python coders) to extract useful data from the PDB website. This will accelerate research in structural biology, drug discovery, and bioinformatics.

## Background: PDB Data Organization and APIs

RCSB.org is powered by publicly available APIs. An API (Application Programming Interface) contains data or features that other applications can access and implement. All static data that is exposed in RCSB.org is available in the Data API. The search API programmatically exposes all search functionality available at rcsb.org.

The RCSB PDB offers a GraphQL API for efficient data retrieval and exploration. GraphQL nodes and edges enable researchers to execute complex queries across data hierarchies using tools like GraphiQL, streamlining relationship traversal and enhancing data integration and analysis capabilities. To use the Data API the user needs to understand PDB data organization.

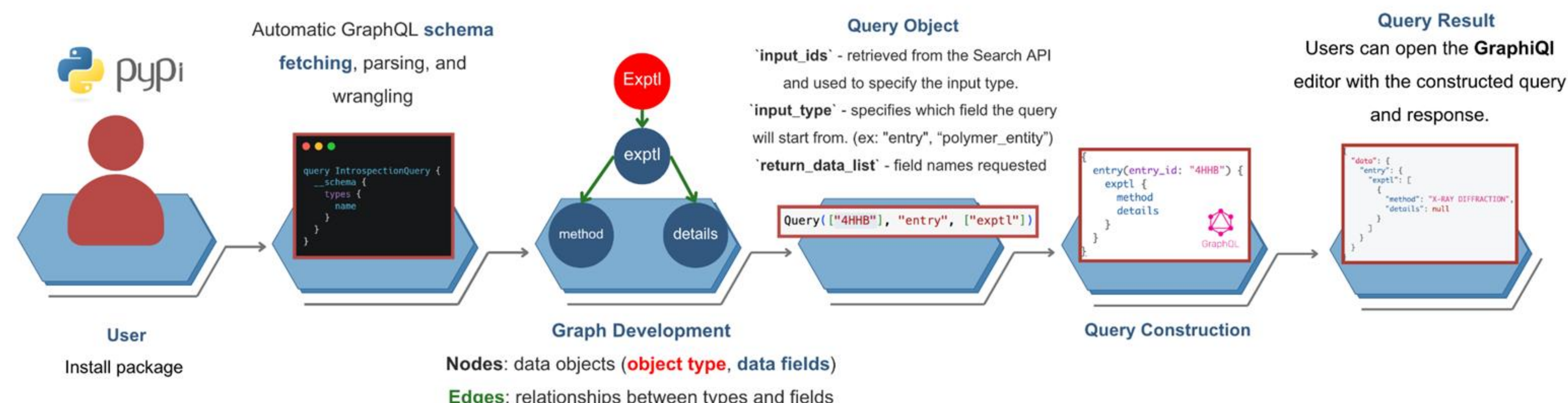


The Protein Data Bank (PDB) uses a hierarchical system to organize 3D macromolecular structure data, spanning from atoms to complete assemblies along with detailed annotations. For example, a deposited ENTRY contains one or more INSTANCES of at least one polymer or branched ENTITY arranged in one or more ASSEMBLIES.

**Given the complexity of PDB data, there is a need for a software package that minimizes user input while delivering comprehensive results.**

## Methods: Python Package Development

Upon initialization, the Python package fetches the requested GraphQL schema, parses the schema, and creates a graph object with over 2500 nodes and edges. This graph representation of how field and type nodes connect helps to automatically create a query using the shortest path algorithm

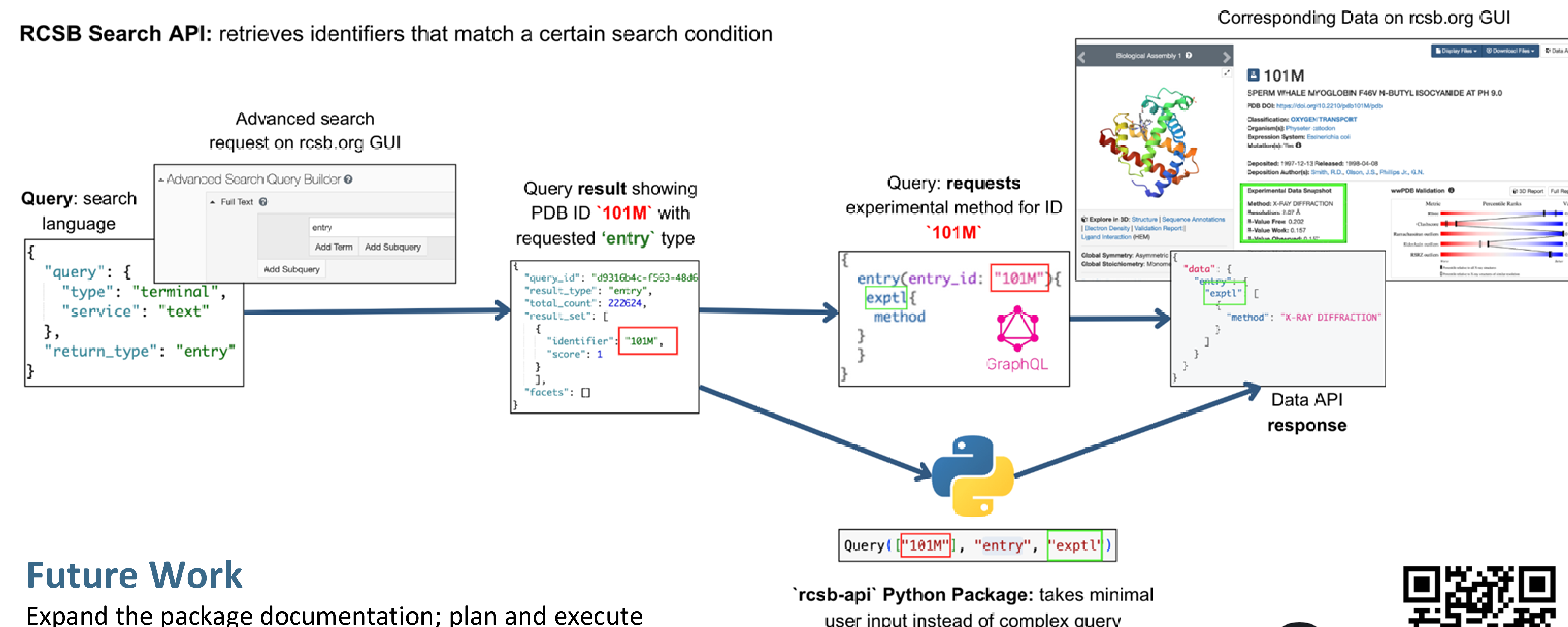


## Results & Conclusion: RCSB PDB API Workflow

The Search API is first used to find PDB IDs that meet specific criteria and types (e.g., entry, assembly). After receiving a response containing the IDs, the `rcsb-api` Python package is used for simplified programmatic access to data about each PDB ID.

**RCSB Data API:** retrieves static data for each structure when you know the PDB identifier

**RCSB Search API:** retrieves identifiers that match a certain search condition



## Future Work

Expand the package documentation; plan and execute the release of new versions of the rcsb-api package to include enhanced features, bug fixes, and performance improvements.

## References:

- [1] Rose, Y., Duarte, J., Lowe, R., Segura, J., Bi, C., et al. (2020). JMB, 433, 166704
- [2] Burley, S., Piehl, D., Vallat, B., Zardecki, C., (2024). IUCrJ, 11, 279-286

